A Complete Derivation of Matter, Antimatter, and Dark Matter in the Harmonic Framework

Author: Peter James Thompson

Date: 21st June 2026

Correspondence: peterjamesthompson101@gmail.com

Parent GUT Reference: Thompson, P.J. (2026). The Harmonic Framework of Reality. Zenodo (deleted by CERN, author's copy available upon request).

Preprint DOI: Not available due to CERN deletion of author's Zenodo account. This paper is part of the public record.

Abstract

The Harmonic Framework of Reality posits three orthogonal time dimensions: T_1 (forward time, matter), T_2 (reverse time, antimatter), and T_3 (orthogonal time, dark matter). This paper presents the complete derivation of the three branches, their mathematical structure, and their physical manifestations.

I show that:

1. **The n-axis belongs exclusively to T_1 (matter).** The dimensional access parameter n = ln(P)/ln(27) describes only the matter branch. It does not apply to T_2 or T_3.
2. **Antimatter (T_2) has no n.** It is accessed through the frequency cutoff f_c2 = 27/230 = 0.1174 Hz. Its mass ladder is m_anti = (h·f_c2/c^2) × k.
3. **Dark matter (T_3) has no n.** It is accessed through the frequency cutoff f_c3 = 27 × (13/19) / 230 = 0.08032 Hz. Its mass ladder is m_dm = (h·f_c3/c^2) × k.
4. **The three time dimensions are orthogonal.** They form the 6D metric $ds^2 = -c^2dt_1^2 - c^2dt_2^2 - c^2dt_3^2 + dx^2 + dy^2 + dz^2$.
5. **The 230th subharmonic is the boundary between T_1 and T_2.** Below f_c2 = 0.1174 Hz, the lattice decoheres from matter and crosses into antimatter.
6. **The 19:13:4 ratio sets the dark matter scale.** The T_3 cutoff is f_c3 = f_c2 × (13/19).
7. **The GUT equation E = m·c²·(v/c)^(n) applies only to T_1.** The unified energy law describes matter only. Antimatter and dark matter require separate equations.
8. **The full harmonic framework unifies all three branches.** Together, they describe the complete structure of reality.

Keywords: Three time dimensions, T_1 matter branch, T_2 antimatter branch, T_3 dark matter branch, dimensional access parameter n, 230th subharmonic, 19:13:4 ratio, orthogonal time, 6D metric, frequency cutoffs

---

1. Introduction: The Three Branches of Time

1.1 The Three Time Dimensions

The Harmonic Framework posits three orthogonal time dimensions:

- **T_1 (forward time):** The matter branch. This is the time we experience.
- **T_2 (reverse time):** The antimatter branch. This is time reversed.
- **T_3 (orthogonal time):** The dark matter branch. This is time orthogonal to both T_1 and T_2.

The 6D metric:

$ds^2 = -c^2dt_1^2 - c^2dt_2^2 - c^2dt_3^2 + dx^2 + dy^2 + dz^2$

1.2 The Key Insight

The n-axis belongs exclusively to T_1 (matter).

The dimensional access parameter $n = \ln(P)/\ln(27)$ describes only the matter branch. It does not apply to T_2 or T_3.

Antimatter and dark matter are not on the n-axis at all.

This is the central insight of this paper.

1.3 Why This Matters

The Standard Model treats antimatter as "negative matter" (negative n). This is a mathematical convenience, not a physical reality.

The Harmonic Framework says: antimatter is not negative matter. It is orthogonal matter.

Dark matter is not complex matter. It is orthogonal to both.

---

2. The Harmonic Framework Recap

2.1 The 27 Hz Anchor

The Tau lattice is anchored at $f_0 = 27$ Hz.

Evidence for the 27 Hz anchor:

Phenomenon	Measured Frequency	Relation to 27 Hz
Schumann 4th mode	27.3 Hz	Within 1.1%
Cat purr fundamental	27-44 Hz	Direct match
Sonoluminescence optimal	26.1-27.9 kHz	×1000
LIPUS modulation	27 Hz, 13.5 Hz	Direct and ×½
Quartz oscillator	32.768 kHz	×1214

2.2 The 19:13:4 Ratio

The 19:13:4 ratio (Marvin's signature: Σ13Δ) appears throughout the framework:

- 19 + 13 = 32 (the number of coherent harmonics)

- $19 + 13 + 4 = 36$, and $36 \times 3/4 = 27$
- $19 - 13 = 6$ (the dimensions of the 6D metric)
- $19/13 \approx 1.4615$ (the generation scaling factor)
- $13/4 = 3.25$ (the second generation scaling factor)

2.3 The 230th Subharmonic

The 230th subharmonic is:

$f\_c2 = 27$ Hz / 230 ≈ 0.1174 Hz

Derivation:

$230 = (19 \times 13) - (13+4) = 247 - 17$

This is the T_1/T_2 boundary.

2.4 The 19:13 Ratio and the T_3 Cutoff

The T_3 cutoff is:

$f\_c3 = 27$ Hz × (13/19) / 230 ≈ 0.08032 Hz

This is the T_3 cutoff frequency.

2.5 The -1° Phase Offset

The echo index Pθ-1 encodes a universal phase offset:

$\varphi_0 = -1° = -\pi/180$ radians

2.6 The Three Time Dimensions

The Tau lattice yields three orthogonal time dimensions:

- T_1: forward time (matter branch)
- T_2: reverse time (antimatter branch)
- T_3: orthogonal time (dark matter branch)

---

3. The Matter Branch (T_1) and the n-Axis

3.1 The Dimensional Access Parameter n

$n = \ln(P)/\ln(27)$

where P is the product of active harmonic indices.

This parameter applies only to T_1 (matter).

3.2 The n-Axis

The n-axis is the axis of the matter branch.

n	Physical Regime	Harmonic Product
1	Momentum	$P = 27$ (k=1)
2	Kinetic Energy	$P = 27\times27$ (k=1,2)
3	Wave→Particle	$P = 27\times54\times81$ (k=1,2,3)
3.54	3D Spatial Volume	$P = 27\times54\times81$
4.54	4D Spacetime	$P = 27\times54\times81\times108$
11.22	Electroweak Scale	$P = 27\times54\times81\times108\times135\times162\times189\times216$
54.65	Planck Scale	$P = 27^{32} \times 32!$

3.3 The Unified Energy Law

For T_1 (matter):

$E = m\cdot c^2\cdot(v/c)^\wedge(n)$

where $n = \ln(P)/\ln(27)$.

This applies only to matter.

3.4 The n-Axis Is Not the Full Story

The n-axis describes matter. It does not describe antimatter or dark matter.

Antimatter and dark matter are orthogonal to the n-axis.

---

4. The Antimatter Branch (T_2)

4.1 T_2 Has No n

Antimatter is not on the n-axis.

It has no n value.

4.2 The T_2 Cutoff Frequency

The T_2 cutoff is:

$f\_c2 = 27$ Hz / 230 = 0.1174 Hz

This is the boundary between T_1 and T_2.

Below this frequency, the lattice decoheres from matter and crosses into antimatter.

4.3 The Antimatter Mass Ladder

The T_2 mass ladder:

$m\_anti = (h\cdot f\_c2/c^2) \times k$

where $k = 1, 2, 3, ...$

Fundamental mass:

$m\_anti\_1 = (h\cdot 0.1174/c^2) \times 1$

$= (6.626\times10^{-34} \times 0.1174) / (3\times10^8)^2$

$= 7.78\times10^{-35} / 9\times10^{16}$

$= 8.64\times10^{-52}$ kg

$= 4.85\times10^{-14}$ eV

The mass ladder:

k	Mass (eV)	Candidate
1	4.85×10^{-14}	Ultralight
10^{12}	0.0485	Sterile neutrino
10^{15}	48.5	Warm antimatter
10^{22}	4.85×10^{8}	WIMP antimatter

4.4 The Antimatter Equation

For T_2 (antimatter):

f_c2 = 27/230 = 0.1174 Hz

m_anti = (h·f_c2/c²) × k

No n appears.

4.5 Antimatter in the 6D Metric

The antimatter branch exists on the t_2 axis of the 6D metric:

$ds^2 = -c^2dt_1^2 - c^2dt_2^2 - c^2dt_3^2 + dx^2 + dy^2 + dz^2$

Antimatter is reverse time.

---

5. The Dark Matter Branch (T_3)

5.1 T_3 Has No n

Dark matter is not on the n-axis.

It has no n value.

5.2 The T_3 Cutoff Frequency

The T_3 cutoff is:

f_c3 = 27 Hz × (13/19) / 230 = 0.08032 Hz

Derivation:

27 × (13/19) = 18.47368

18.47368 / 230 = 0.08032 Hz

This is the T_3 cutoff frequency.

5.3 The Dark Matter Mass Ladder

The T_3 mass ladder:

m_dm = (h·f_c3/c²) × k

where k = 1, 2, 3, ...

Fundamental mass:

m_dm_1 = $(h \cdot 0.08032/c^2) \times 1$

$= (6.626\times10^{-34} \times 0.08032) / (3\times10^8)^2$

$= 5.32\times10^{-35} / 9\times10^{16}$

$= 5.91\times10^{-52}$ kg

$= 3.31\times10^{-14}$ eV

The mass ladder:

k	Mass (eV)	Candidate
1	3.31×10^{-14}	Ultralight
10^{12}	0.0331	Sterile neutrino
10^{15}	33.1	Warm dark matter
10^{22}	3.31×10^{8}	WIMP dark matter

5.4 The Dark Matter Equation

For T_3 (dark matter):

f_c3 = 27 × (13/19) / 230 = 0.08032 Hz

m_dm = $(h \cdot f\_c3/c^2) \times k$

No n appears.

5.5 Dark Matter in the 6D Metric

The dark matter branch exists on the t_3 axis of the 6D metric:

$ds^2 = -c^2dt_1^2 - c^2dt_2^2 - c^2dt_3^2 + dx^2 + dy^2 + dz^2$

Dark matter is orthogonal time.

---

6. Comparison of the Three Branches

6.1 The Complete Table

Branch	Time Dimension	n	Equation	Cutoff Frequency
Matter	T_1	Yes (1, 2, 3, ..., 54.65)	$E = m \cdot c^2 \cdot (v/c)^{(n)}$	None (continuous)
Antimatter	T_2	No	m_anti = $(h \cdot f\_c2/c^2) \times k$	f_c2 = 27/230
Dark Matter	T_3	No	m_dm = $(h \cdot f\_c3/c^2) \times k$	f_c3 = 27×(13/19)/230

6.2 The Cutoff Frequencies

T_2 cutoff:

f_c2 = 27/230 = 0.1174 Hz

T_3 cutoff:

$f_{c3} = 27 \times (13/19) / 230 = 0.08032$ Hz

The ratio:

$f_{c3} / f_{c2} = (13/19) = 0.6842$

6.3 The Mass Ladders

T_2 mass ladder:

$m_{anti} = (h \cdot f_{c2}/c^2) \times k$

T_3 mass ladder:

$m_{dm} = (h \cdot f_{c3}/c^2) \times k$

T_1 mass ladder:

$m_{matter} = (h \cdot f_0/c^2) \times (P/230) \times k$

where P is the product of harmonic indices.

---

7. The 6D Metric and Orthogonality

7.1 The 6D Spacetime

The full metric is:

$$ds^2 = -c^2dt_1^2 - c^2dt_2^2 - c^2dt_3^2 + dx^2 + dy^2 + dz^2$$

7.2 Orthogonality

The three time dimensions are orthogonal:

$\langle dt_1, dt_2 \rangle = 0$
$\langle dt_1, dt_3 \rangle = 0$
$\langle dt_2, dt_3 \rangle = 0$

This is why T_2 and T_3 have no n.

The n-axis is the axis of T_1.

T_2 and T_3 are orthogonal to T_1 and to each other.

7.3 The Projection

We observe only the projection onto T_1.

The observed universe is the projection of the 6D metric onto the T_1 axis.

---

8. The 230th Subharmonic as the T_1/T_2 Boundary

8.1 The Boundary

The 230th subharmonic is the boundary between T_1 and T_2.

f_c2 = 27/230 = 0.1174 Hz

Below this frequency, the lattice decoheres from matter and crosses into antimatter.

8.2 Why 230?

230 = (19×13) - (13+4) = 247 - 17

8.3 The Decoherence

At f_c2 = 0.1174 Hz:

1. The lattice can no longer maintain coherent matter-branch oscillations.
2. The n-axis crosses from positive (T_1) to a state where n no longer applies.
3. The system crosses into T_2 (antimatter).

This is the T_1/T_2 boundary.

---

9. The 19:13:4 Ratio as the Dark Matter Scale

9.1 The T_3 Cutoff

The T_3 cutoff is:

f_c3 = f_c2 × (13/19) = 0.1174 × 0.6842 = 0.08032 Hz

9.2 Why 13/19?

The 13 in the 19:13:4 ratio is the consciousness anchor.

The 19 in the 19:13:4 ratio is the matter anchor.

The T_3 branch is accessed through the consciousness ratio (13/19).

9.3 The Full Hierarchy

Level	Ratio	Frequency	Branch
1	19	f_c2 = 0.1174 Hz	T_2 (antimatter)
2	13	f_c3 = 0.08032 Hz	T_3 (dark matter)
3	4	Redundancy	Error correction

---

10. The Complete Hierarchy of Reality

10.1 The n-Axis (T_1 Only)

```
        n-axis (T₁ only)
            ↑
            |
...         |
54.65  ---+---  Planck Scale (matter branch decoheres)
11.22  ---+---  Electroweak (matter)
4.54   ---+---  4D Spacetime (matter)
3.54   ---+---  Wave→Particle (matter)
2      ---+---  Kinetic Energy (matter)
1      ---+---  Momentum (matter)
0      ---+---  T₁/T₂ Boundary (f_c2 = 0.1174 Hz)
```

10.2 The T_2 and T_3 Axes

```
        n-axis (T₁ only)
            ↑
            |
            |
            |
            |
            |
            |
            |          T₂-axis
            |          (f_c2 = 0.1174 Hz)
            |          (m_anti = (h·f_c2/c²) × k)
            |
            |          T₃-axis
            |          (f_c3 = 0.08032 Hz)
            |          (m_dm = (h·f_c3/c²) × k)
```

10.3 The Full 6D Structure

```
            T₁ (n-axis)
            ↑
            |
            |          T₂ (orthogonal)
            |          ↑
            |          |
            |          |          T₃ (orthogonal)
            |          |          ↑
            |          |          |
            +----------+----------+---→ Space
```

11. Testable Predictions

11.1 The T_2 Cutoff

Prediction 1: The T_2 cutoff frequency is $f_{c2} = 0.1174$ Hz.

Test: Search for a cutoff at 0.1174 Hz in cosmological data.

11.2 The T_3 Cutoff

Prediction 2: The T_3 cutoff frequency is $f_{c3} = 0.08032$ Hz.

Test: Search for a cutoff at 0.08032 Hz in dark matter detector data.

11.3 The Antimatter Mass Ladder

Prediction 3: The antimatter mass ladder is $m_{anti} = (h \cdot f_{c2}/c^2) \times k$.

Test: Search for antimatter candidates at masses corresponding to $k = 1, 2, 3, \ldots$

11.4 The Dark Matter Mass Ladder

Prediction 4: The dark matter mass ladder is $m_{dm} = (h \cdot f_{c3}/c^2) \times k$.

Test: Search for dark matter candidates at masses corresponding to $k = 1, 2, 3, \ldots$

11.5 The 19:13:4 Ratio

Prediction 5: The 19:13:4 ratio should appear in all three branches.

Test: Look for the 19:13:4 ratio in matter, antimatter, and dark matter spectra.

---

12. Conclusion

12.1 Summary

The three branches of time are:

1. **T_1 (matter):** Forward time, accessed through $n = \ln(P)/\ln(27)$.
2. **T_2 (antimatter):** Reverse time, accessed through $f_{c2} = 27/230 = 0.1174$ Hz.
3. **T_3 (dark matter):** Orthogonal time, accessed through $f_{c3} = 27\times(13/19)/230 = 0.08032$ Hz.

The n-axis belongs exclusively to T_1.

T_2 and T_3 have no n.

They are accessed through frequency cutoffs, not through n.

12.2 The Equations

For T_1 (matter):

$n = \ln(P)/\ln(27)$
$E = m \cdot c^2 \cdot (v/c)^{(n)}$

For T_2 (antimatter):

$f\_c2 = 27/230 = 0.1174$ Hz
$m\_anti = (h \cdot f\_c2/c^2) \times k$

For T_3 (dark matter):

$f\_c3 = 27 \times (13/19) / 230 = 0.08032$ Hz
$m\_dm = (h \cdot f\_c3/c^2) \times k$

12.3 The Physical Meaning

The universe is not just matter. It is matter (T_1), antimatter (T_2), and dark matter (T_3) existing in three orthogonal time dimensions.

The stones are in the pond. The ripples are in three directions we cannot see.

We only see one.

12.4 The Final Word

The stones are in the pond. The 27 Hz anchor is the stone. The three time dimensions are the ripples moving in three orthogonal directions.

The universe is not one-dimensional in time. It is three-dimensional.

We are just learning to see the other two.

References

[1] Thompson, P.J. (2026). The Harmonic Framework of Reality: A Complete Grand Unified Field Theory.

[2] Thompson, P.J. (2026). "The Full 32-Harmonic Chords of the Standard Model." Preprint.

[3] Thompson, P.J. (2026). "The T_3 Dark Matter Dimension: A Complete Derivation." Preprint.

[4] Thompson, P.J. (2026). "The 230th Subharmonic as Natural Ultraviolet Cutoff: A Finite Quantum Field Theory." Preprint.

[5] Thompson, P.J. (2026). "The SPS Ghost Resonance as Direct Evidence of the Tau Lattice." Preprint.

[6] Thompson, P.J. (2026). "The Hubble Tension as a Harmonic Phase Effect in the Tau Lattice." Preprint.

[7] Thompson, P.J. (2026). "Neutrino Mass Hierarchy from Harmonic Chords and the -1° Phase Offset." Preprint.

[8] Thompson, P.J. (2026). "The Cosmic Microwave Background as a Fossilised Tau Interference Pattern." Preprint.

[9] Thompson, P.J. (2026). "Astrophysical Jets as Harmonic Resonances in the Tau Lattice." Preprint.

[10] Thompson, P.J. (2026). "Topological Quantum Field Theory as a Dual of the Tau Lattice." Preprint.

[11] Thompson, P.J. (2026). "Quantum Entanglement as Harmonic Coherence Across T_2." Preprint.

[12] Thompson, P.J. (2026). "Quantum Gravity as the Gradient of the 27 Hz Anchor." Preprint.

[13] Thompson, P.J. (2026). "Gamma-Ray Bursts as Harmonic Collapse Events." Preprint.

[14] Thompson, P.J. (2026). "Fast Radio Bursts as Tau Lattice Nodes." Preprint.

[15] Thompson, P.J. (2026). "The Black Hole Information Paradox as Phase Memory." Preprint.

[16] Thompson, P.J. (2026). "Time Crystals and the 27 Hz Anchor." Preprint.

[17] Thompson, P.J. (2026). "Quantum Zeno Effect as Phase Locking." Preprint.

[18] Thompson, P.J. (2026). "The Anthropic Principle as Harmonic Selection." Preprint.

[19] Thompson, P.J. (2026). "Dark Energy as Lattice Tension." Preprint.

[20] Thompson, P.J. (2026). "The Harmonic Vampire: Phase Entrainment and Consciousness Drain at 13.04 Hz." Preprint.

Correspondence: peterjamesthompson101@gmail.com

Supplementary material: Complete code, simulations, and blueprints available at https://github.com/peterjamesthompson101-svg/Physics-Papers